

DO NOT WRITE ON TEST



**INFORMATION
TECHNOLOGY CONCEPTS
~OPEN EVENT~
(391)
STATE 2023**

TOTAL POINTS (50 @ 2 points each) _____ **100 Points**

Test Time: 60 minutes

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. What does USB stand for when referring to a USB cable?
 - A. Unified Serial Bit
 - B. Universal Standard Bit
 - C. Universal Serial Bus
 - D. Unified Standard Bus
2. What does RAM stand for when referring to computer RAM?
 - A. Reserved Accessible Memory
 - B. Random Access Memory
 - C. Reserved Automatic Memory
 - D. Random Automatic Movement
3. What does CPU stand for when referring to a computer CPU?
 - A. Central Processing Unit
 - B. Component Processor Unit
 - C. Computer Programmed Unit
 - D. Component Programming Unit
4. What does GPU stand for when referring to a computer GPU?
 - A. Graphics Processing Unit
 - B. Grader Processor Unit
 - C. Graphics Programmed Unit
 - D. Graphical-Interface Programming Unit
5. In programming, a single character data type with the size of one byte, can be represented by?
 - A. String
 - B. int
 - C. double
 - D. char
6. Which of the following contains data amounts listed from smallest to largest?
 - A. Kilobyte, Megabyte, Gigabyte, Terabyte
 - B. Megabyte, Kilobyte, Terabyte, Gigabyte
 - C. Gigabyte, Terabyte, Kilobyte, Megabyte,
 - D. Kilobyte, Gigabyte, Megabyte, Terabyte

7. Which of the following is NOT a software program?
 - A. Microsoft Word
 - B. Adobe Illustrator
 - C. Hard Disk Drive
 - D. Google Chrome
8. What cable is used for providing an internet connection?
 - A. HDMI
 - B. Audio jack
 - C. Fiber optic
 - D. DP
9. Audio cables are typically labeled with a color designation. What color does pink represent?
 - A. Audio out
 - B. Audio in
 - C. Microphone in
 - D. Microphone out
10. How many bytes are in 1 Kilobyte (kB)?
 - A. 256 Bytes
 - B. 512 Bytes
 - C. 1,024 Bytes
 - D. 2,048 Bytes
11. What is the primary responsibility of a network switch?
 - A. Turn power on and off to a room
 - B. Connect devices to a LAN
 - C. Connect devices to global cross-country network
 - D. Provide Wi-Fi signal to devices within 50 meters
12. A firewall uses an ____ that regulates traffic entering and exiting networks?
 - A. Bitlocker
 - B. Encrypted Playback Manager
 - C. Access Control List
 - D. Controlled Environment Manager
13. MAC addresses for a computer are a ____ hexadecimal address.
 - A. 8 bit
 - B. 16 bit
 - C. 24 bit
 - D. 48 bit

14. Which port is primarily used for HTTPS services?
- A. 80
 - B. 160
 - C. 213
 - D. 443
15. Which port is primarily used for HTTP services?
- A. 80
 - B. 160
 - C. 213
 - D. 443
16. The three main components of cybersecurity have the acronym CIA. What does CIA stand for?
- A. Confidentiality, Integrity, Availability
 - B. Confidentiality, Interference, Angle
 - C. Conspire, Inspire, Aspire
 - D. Confidentiality, Interference, Availability
17. A technician is troubleshooting a problem. The tech tests the theory and determines the theory is confirmed. Which of the following should be the technician's next step?
- A. Implement the solution
 - B. Document lessons learned
 - C. Establish a plan of action
 - D. Verify full system functionality
18. Which of the following is a threat to confidentiality?
- A. Replay attack
 - B. Denial of service
 - C. Service outage
 - D. Dumpster diving
19. A systems administrator wants to return results for time range within a database. Which of the following commands should the administrator use?
- A. INSERT
 - B. SELECT
 - C. DELETE
 - D. UPDATE

20. Without the need of help desk, users can reset their passwords themselves using what service?
- A. Self-Service Password Creator
 - B. Self-Service Portal
 - C. Self-Service Password Reset
 - D. Self-Service System
21. Which of the following computing devices would be used to provide a centralized means to distribute services to a group of clients and usually possesses a role on a LAN?
- A. Laptop
 - B. Server
 - C. Workstation
 - D. Mobile phone
22. A user, Ann, is concerned about theft of her laptop and does not want a thief to have easy access to all of her banking and email. Which of the following precautions could be taken to mitigate this issue?
- A. Only browse the Internet on WiFi connections that use WPA2
 - B. Turn off the guest account in the operating system
 - C. Remove any legacy browsers from the computer
 - D. Disable autofill functionality within the web browser
23. A user is configuring a new wireless router. Which of the following should be done to ensure that unauthorized changes cannot be made?
- A. Change the admin password
 - B. Change the SSID
 - C. Change the router's address
 - D. Change the encryption key
24. Malware that has an embedded keylogger to capture all of the keystrokes and steal logins is considered what?
- A. Adware
 - B. Ransomware
 - C. Phishing
 - D. Spyware
25. A user needs to download tax documents from a financial website. Which of the following is the website most likely to use for transmission of the tax document to the user's browser?
- A. HTTPS
 - B. HTTP
 - C. SFTP
 - D. FTP

26. Which of the following is an advantage of using cloud-based collaborative applications and storage, rather than local applications and storage?
- A. Decreased software licensing costs
 - B. Higher security encryption
 - C. Limited storage space for files
 - D. Increased accessibility to files
27. A technician is configuring a wireless router for a small office and the business owner would like the wireless network to be secured using the strongest encryption possible. Which of the following the technician choose?
- A. WAP
 - B. WPA
 - C. WPA2
 - D. WEP
28. Which of the following permissions is required to run a .bat file?
- A. Execute
 - B. Write
 - C. Modify
 - D. Delete
29. A home computer that has been in operation for more than five years is now suffering from random shutdowns during the day. The PC is set up in an enclosed cubby built into the home office desk. Which of the following is most likely plaguing the PC?
- A. The PC was switched over to 240v operation mode
 - B. The PC' BIOS is severely out-of-date, causing PSU issues
 - C. EMI
 - D. Dust accumulation
30. If the limit has not been changed from the default value, Windows limits the length of the characters that are allowed in a file name. What is the limit?
- A. 32
 - B. 64
 - C. 130
 - D. 260
31. A scripting language is used to give software specific commands. Which of the following is a scripting language?
- A. C
 - B. C++
 - C. COBOL
 - D. Python

32. A cyber-attack just occurred on a resource you were using. It appears multiple computers were used to attack your resource. What type of attack just occurred?
- A. DDD
 - B. DDoS
 - C. DoS
 - D. DDoSS
33. Which of the following languages would a web developer most likely use to create a website?
- A. Interpreted
 - B. Query
 - C. Compiled
 - D. Assembly
34. Which of the following would be best to keep data on a laptop safe if the laptop is lost or stolen?
- A. Host-based firewall
 - B. Strong admin password
 - C. Anti-malware software
 - D. Full disk encryption
35. Which one of the following will most likely be used when sending and receiving Internet email?
- A. POP3
 - B. DHCP
 - C. ICMP
 - D. SFTP
36. Which of the following best describes an application running on a typical operating system?
- A. Process
 - B. Thread
 - C. Function
 - D. Task
37. Which of the following software license models allows a developer to modify the original code and release its own version of the application?
- A. Proprietary software
 - B. Commercial software
 - C. Open-source software
 - D. Cross-platform software

38. Which of the following components provides the fastest medium for hosting and storing an operating system?
- A. SSD
 - B. Blu-ray disc
 - C. 7200 RPM HDD
 - D. DVD-RW
39. Which one of the following is an example of a keyboard connector?
- A. RJ-11
 - B. Serial
 - C. FireWire
 - D. PS/2
40. Which of the following CPU cooling types is best in a dusty environment?
- A. Liquid cooling
 - B. Open case
 - C. Case fan
 - D. Heat sink fan
41. Which of the following ethernet physical media would support the use of a fiber optic connection?
- A. An ethernet cable that supports a data transmission rate of 100 Mbps
 - B. A networking cable that supports a data transmission rate of 1,000 Kbps
 - C. An ethernet cable that is designed to a 1,000 Mbps specification
 - D. A networking cable that supports a data transmission rate of 500 Kbps
42. A technician is troubleshooting a laptop. He decides to boot the computer into VGA mode from the ABOM. What is the default display resolution for this mode?
- A. 1280 x 800
 - B. 1366 x 768
 - C. 640 x 480
 - D. 1920 x 1080
43. Which of the following best describes the pros and cons of home wired network?
- A. Low throughput, low mobility, high availability, high latency, high security
 - B. High throughput, high mobility, low availability, low latency, high security
 - C. High throughput, low mobility, high availability, low latency, high security
 - D. Low throughput, high mobility, low availability, high latency, low security

44. A technician has just finished setting up a SOHO wireless router but the client does not want the PC on the wireless network for security reasons. The technician connects an RJ-45 cable to the computer and the router, but does not receive network connectivity. Which of the following is the next step to diagnose the problem?
- A. Change the IP address on the computer to match the router
 - B. Check for link light activity on the computer and router
 - C. Reseat the power cable on the computer and reboot
 - D. Change the hostname of the router to match computer's network
45. A user's laptop hard drive contains sensitive information. The user often plugs the laptop into the corporate network. A sensitive file from the laptop has been found on another user's laptop. How could the user have prevented this breach?
- A. Delete unused drives from network
 - B. Remove shared keys from the key ring
 - C. Set the read-only attribute on the files
 - D. Disable file and print sharing on the laptop
46. The LAN in large workplace needs to be set up using the most common cabling and wiring standard. What is the name of this TIA/EIA network standard?
- A. ANSI
 - B. T568A
 - C. T568B
 - D. 802.3af
47. A maintenance technician has recently installed a duplexing assembly into an office laser printer. What does the duplexing assembly enable the printer to do?
- A. A duplexing device enables a black-and-white laser printer to produce color printouts
 - B. High-speed data transmission between a network computer and a laser printer is permitted via the use of a duplexing assembly installed in the laser printer, thereby facilitating comparatively faster printing
 - C. A duplexing assembly enables a laser printer to print on both sides of paper
 - D. Print copies to two different trays
48. Which of the following IEEE Wi-Fi standards are most likely to be affected by the activity of wireless phones, Bluetooth devices, and microwaves?
- A. 802.11a
 - B. 802.11a and 802.11b
 - C. 802.11b and 802.11g
 - D. 802.11b, 802.11g, and 802.11n

49. A customer has informed their local PC repair shop that their flat panel LCD monitor has several faulty pixels. The customer complains that the faulty pixels are affecting her concentration as she can see them in her peripheral visual field as she works on an important project. Which pixel color will most likely require the LCD display to be replaced?
- A. Black
 - B. Blue
 - C. Green
 - D. Yellow
50. Which unit of measurement is typically used to demonstrate the effectiveness and throughput speed of computer air cooling systems?
- A. CDMA
 - B. CFM
 - C. DTR
 - D. TDP